

# COUPLING OF PLASMONS TO THE TWO-MAGNON CONTINUUM IN ANTIFERROMAGNETS

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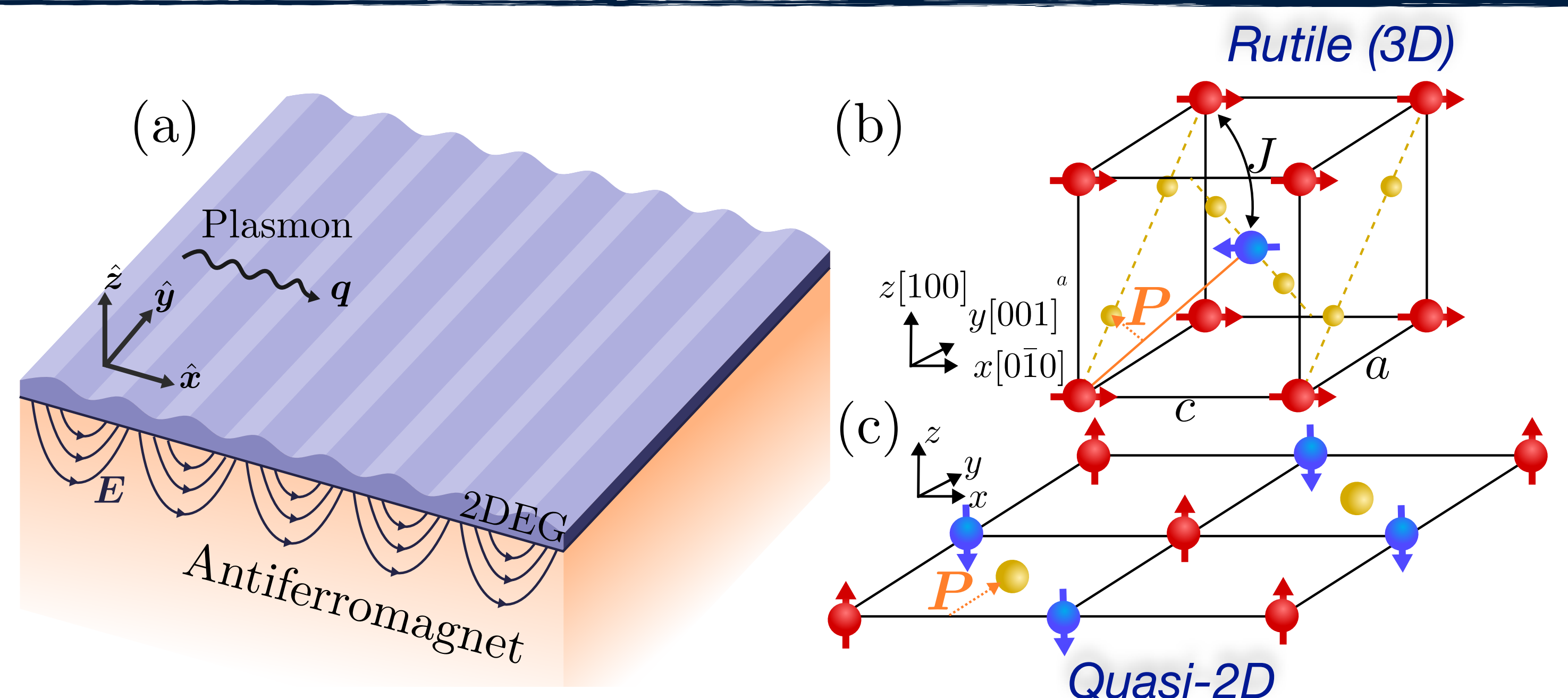


## Setup

Gapless plasmons in a 2DEG on top of an antiferromagnet: electric field extends into the AFM

AFM with a finite polarisation vector  $\rightarrow$  two-magnon continuum couples to electric fields

What are the signatures of the two-magnon continuum on the plasmon dynamics?



## Magnon-plasmon coupling

T. Moriya, J. Phys. Soc. Jpn 23, 490 (1967).

Plasmon electric field  $\hat{E}_i$  couples to the polarisation operator  $\hat{P}_i$ .  $\Rightarrow H_c = - \sum_i \hat{E}_i \cdot \hat{P}_i$

Polarisation operator in an AFM (exchange-striction mechanism):

$$\hat{P}_i = \sum_j \pi_{ij} \hat{S}_i \cdot \hat{S}_j$$

is only non-zero for a bond where its center is not a center of inversion

Introduce Holstein-Primakoff operators for the spins  $\hat{S}_i$

B. A. Ferreira, PRA 101, 033817 (2020).

Plasmon electric field is given by:  $\hat{E}(\mathbf{r}_i) = \frac{1}{\sqrt{N_p}} \sum_q i E_q e^{iq \cdot \mathbf{r}_i - |z_i|/\lambda_q} \hat{\phi}_q + \text{h.c.},$

Plasmon annihilation/creation operator

$$H_c^{(2)} = -iS \frac{1}{\sqrt{N_x N_y}} \sum_n e^{-|z_n|/\lambda_q} \sum_{\mathbf{k} \mathbf{q}} (\hat{\phi}_{\mathbf{q}} E_{\mathbf{q}} - \hat{\phi}_{-\mathbf{q}}^\dagger E_{-\mathbf{q}}^*) \cdot \hat{V}_{\mathbf{k}, \mathbf{q}},$$

Exponential decay of plasmon field

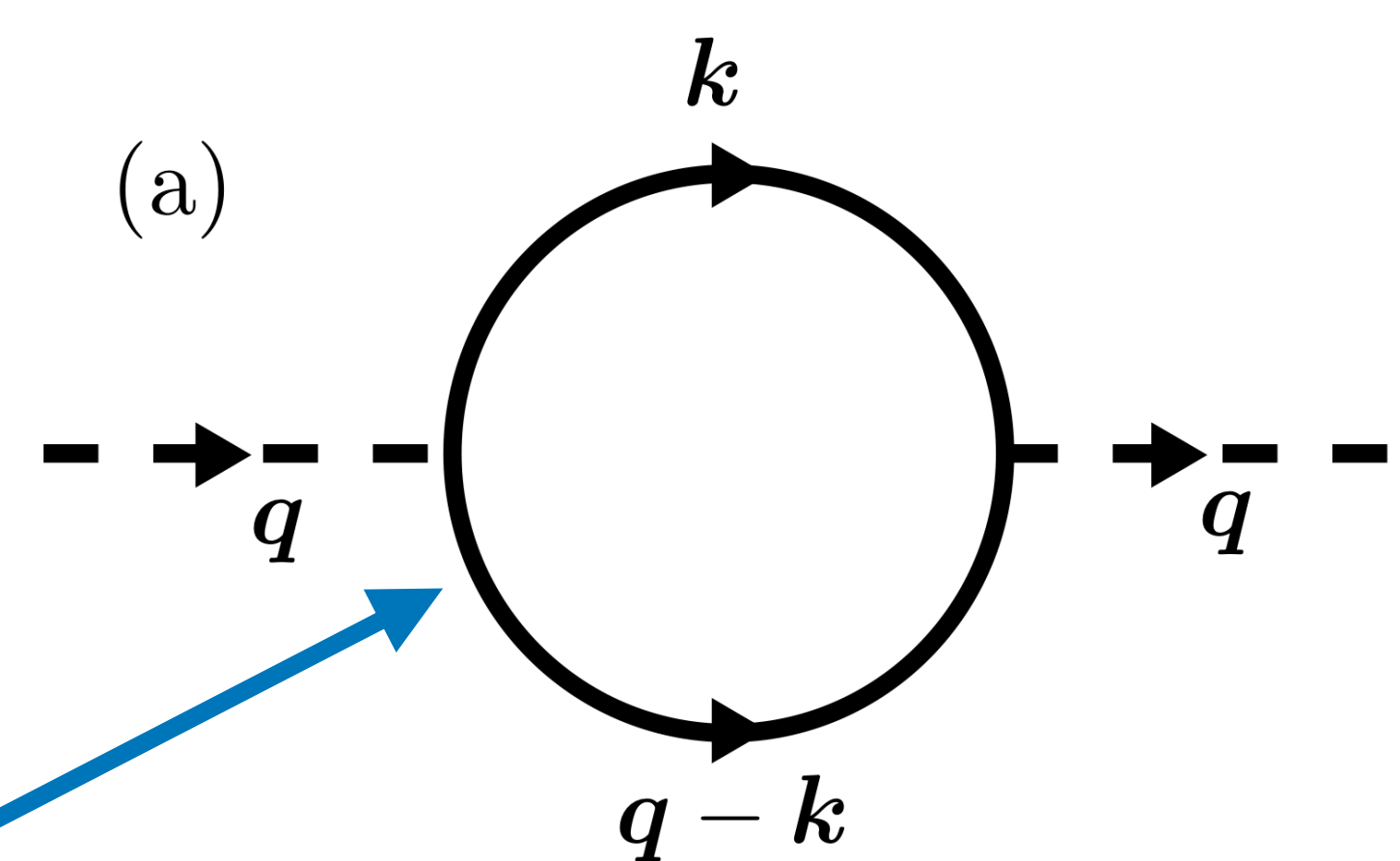
$$\begin{aligned} & \equiv u_{\mathbf{k}} u_{\mathbf{k}-\mathbf{q}} \pi_{-\mathbf{q}-\mathbf{k}}^* \hat{\alpha}_{\mathbf{k}} \hat{\beta}_{-\mathbf{k}-\mathbf{q}} \\ & + u_{\mathbf{k}} u_{\mathbf{k}+\mathbf{q}} \pi_{\mathbf{q}-\mathbf{k}} \hat{\alpha}_{\mathbf{k}}^\dagger \hat{\beta}_{\mathbf{q}-\mathbf{k}}^\dagger \\ & + v_{\mathbf{k}} v_{\mathbf{k}-\mathbf{q}} \pi_{\mathbf{q}+\mathbf{k}} \hat{\alpha}_{-\mathbf{k}-\mathbf{q}} \hat{\beta}_{\mathbf{k}} \\ & + v_{\mathbf{k}} v_{\mathbf{k}+\mathbf{q}} \pi_{\mathbf{k}-\mathbf{q}}^* \hat{\beta}_{\mathbf{k}}^\dagger \hat{\alpha}_{\mathbf{q}-\mathbf{k}}^\dagger \end{aligned}$$

Bogoliubov factors

Magnon annihilation/creation operators

## Plasmon self-energy

Dominant plasmon self-energy because of plasmon-magnon interactions at zero temperature:



$$\Sigma_q^f(\epsilon) = \frac{1}{N_x N_y} \sum_n e^{-|z_n|/\lambda_q} \sum_{\mathbf{k}} \sum_{\sigma, \sigma' = \pm} \frac{1}{2} \frac{|E_{\mathbf{q}} \cdot \mathbf{V}_{\mathbf{q}, \mathbf{k}}^{\sigma \sigma'}|^2}{\epsilon + i0^+ - \epsilon_{\mathbf{k}} - \epsilon_{\mathbf{q}-\mathbf{k}}}$$

Directly related to the two-magnon continuum, which describes the simultaneous creation of two magnons with momentum  $\mathbf{q}$  and energy  $\epsilon$ :

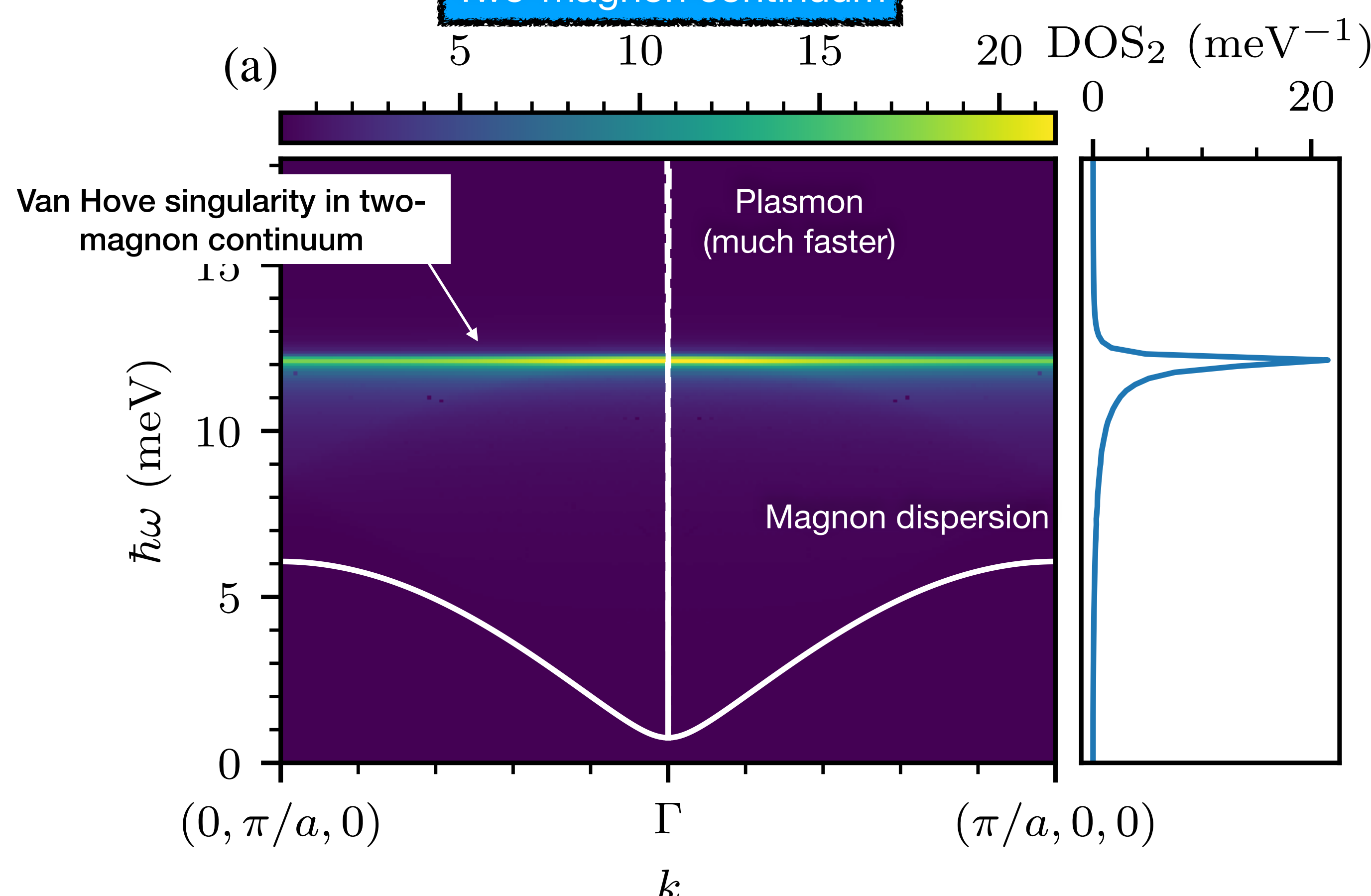
$$D_q(\epsilon) = \frac{1}{N_m} \sum_{\mathbf{k}} \delta(\epsilon - \epsilon_{\mathbf{k}} - \epsilon_{\mathbf{q}-\mathbf{k}}),$$

Therefore: plasmons couple to the two-magnon continuum in antiferromagnets

## Resulting coupling

Results for Rutile-AFM (3D), but quasi-2D similar

Two-magnon continuum



Plasmon spectral density

R. Verresen, Nat. Phys. 15, 750 (2019).

